Sampling

Chapter 8

Mohamed Bingabr

Chapter Outline

- Sampling
- Quantization
- Binary Representation

Sampling Theorem

A real signal whose spectrum is bandlimited to B Hz [$X(\omega) = 0$ for $|\omega| > 2\pi B$] can be reconstructed exactly from its samples taken uniformly at a rate $f_s > 2B$ samples per second. When $f_s = 2B$ then f_s is the Nyquist rate.

$$\bar{x}(t) = x(nT) = x(t)\sum_{n=-\infty}^{n=\infty}\delta(t-nT)$$

$$\bar{x}(t) = x(nT) = x(t)\frac{1}{T}\sum_{n=-\infty}^{n=\infty}e^{jn\omega_s t}$$

$$\bar{X}(\omega) = \frac{1}{T}\sum_{n=-\infty}^{n=\infty}X(\omega - n\omega_s)$$

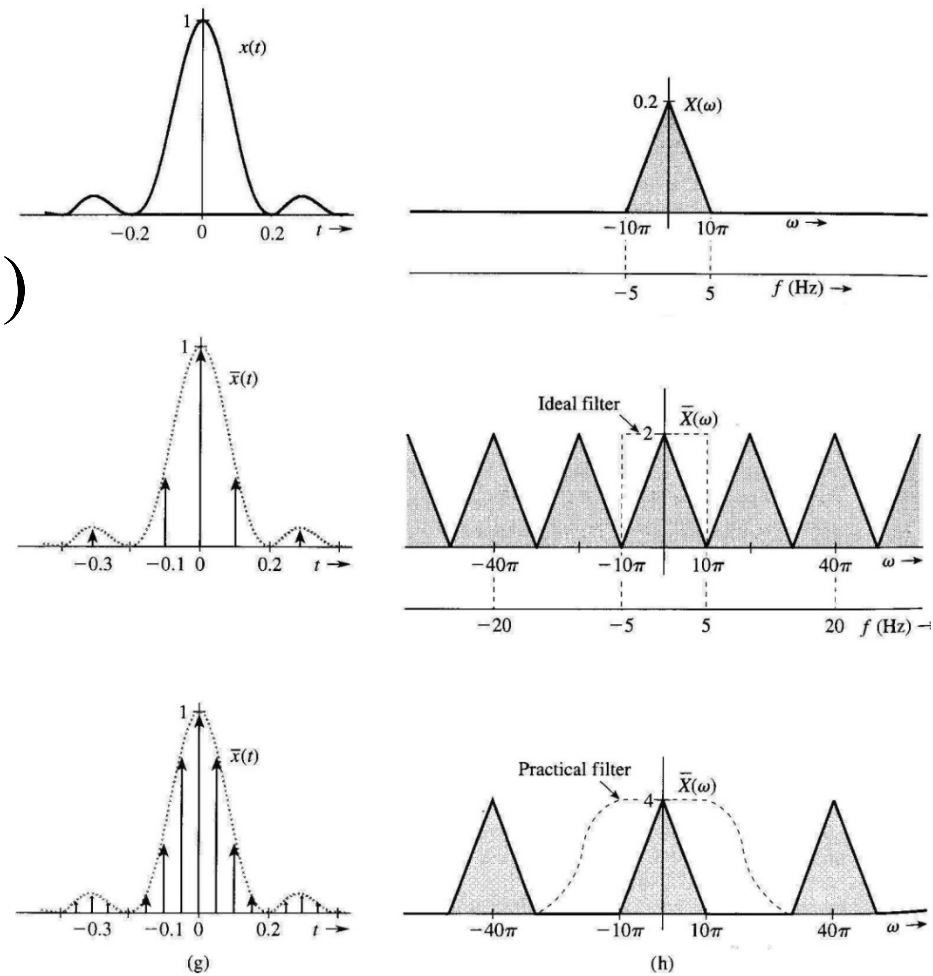
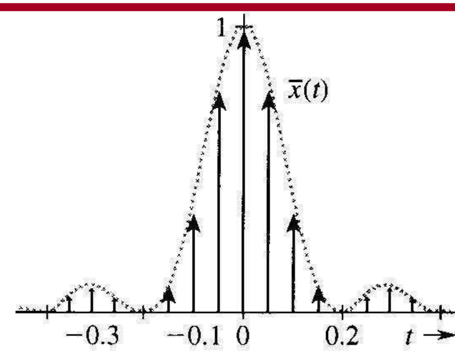
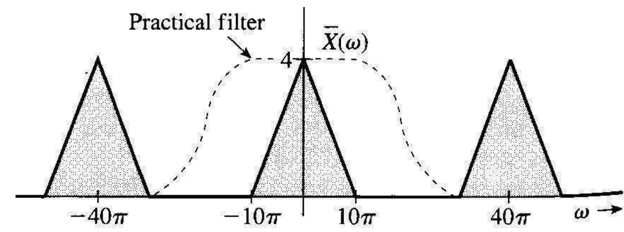


Figure 8.2 Effects of undersampling and oversampling.

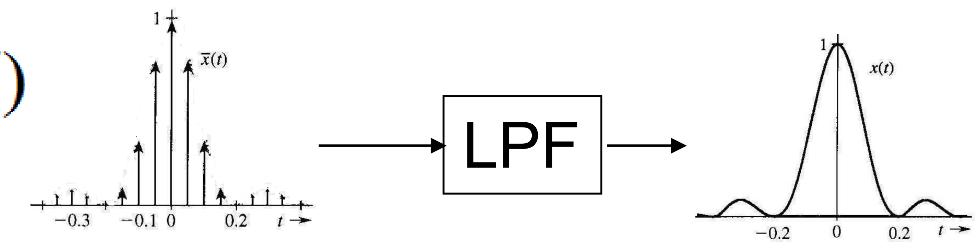
Reconstructing the Signal from the Samples

$$X(\omega) = H(\omega)\,\overline{X}(\omega)$$

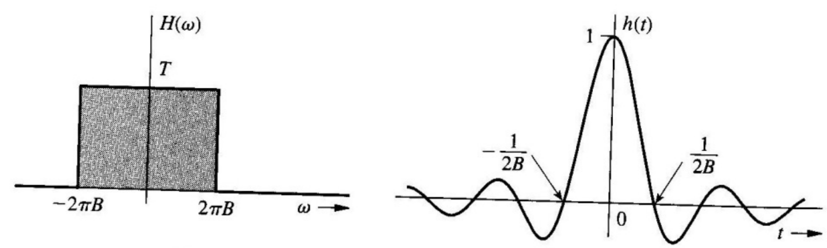




$$x(t) = h(t) * \sum_n x(nT)\delta(t - nT)$$

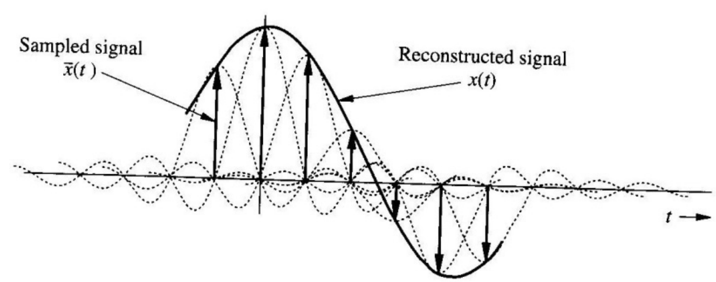


$$x(t) = \sum_n x(nT)h(t - nT)$$



(a) (b)

$$x(t) = \sum_n x(nT)\,\text{sinc}(2\pi B(t - nT))$$



Example

Determine the Nyquist sampling rate for the signal

$$x(t) = 3 + 2\cos(10\pi t) + \sin(30\pi t).$$

Solution

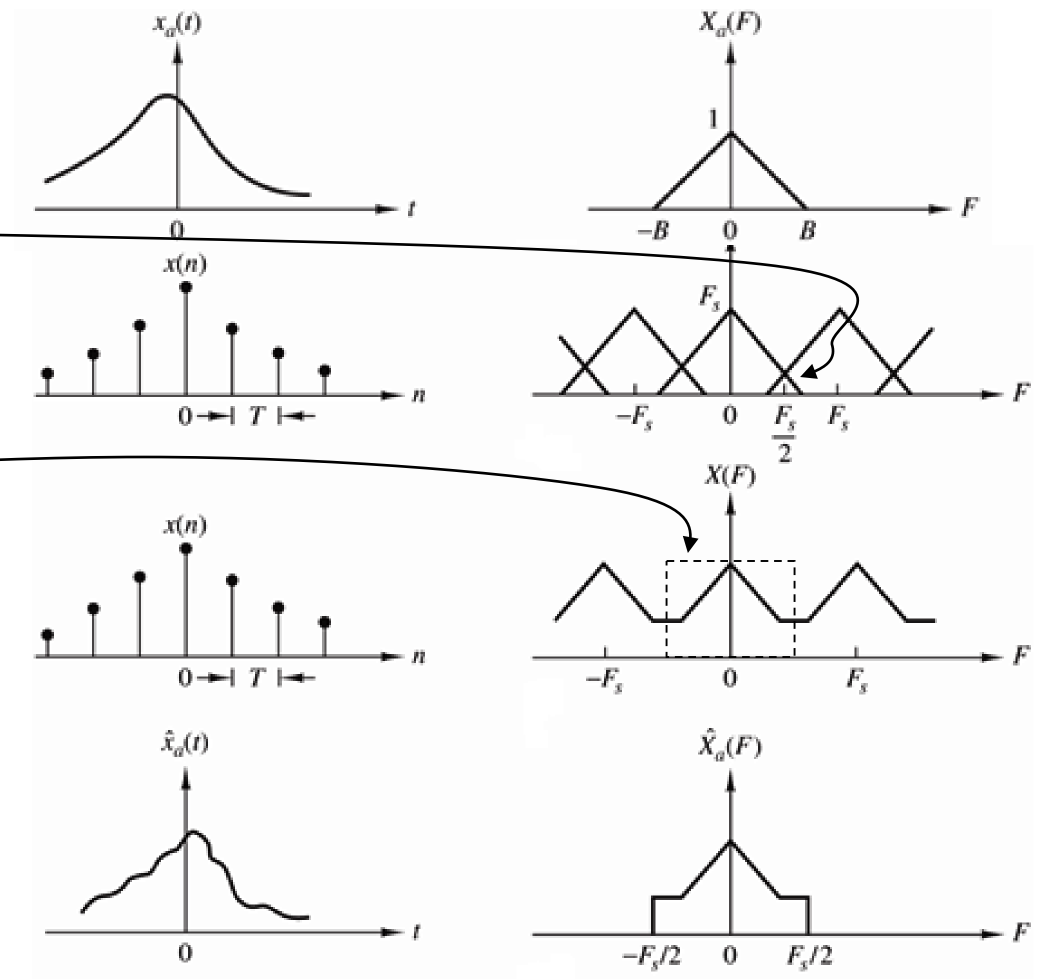
The highest frequency is f_{max} = 30π/2π = 15 Hz
The Nyquist rate = 2 f_{max} = 2*15 = 30 sample/sec

Aliasing

If a continuous time signal is sampled below the Nyquist rate then some of the high frequencies will appear as low frequencies and the original signal can not be recovered from the samples.

Frequency above $f_s/2$ will appear (aliased) as frequency below $f_s/2$

LPF
With cutoff frequency $f_s/2$



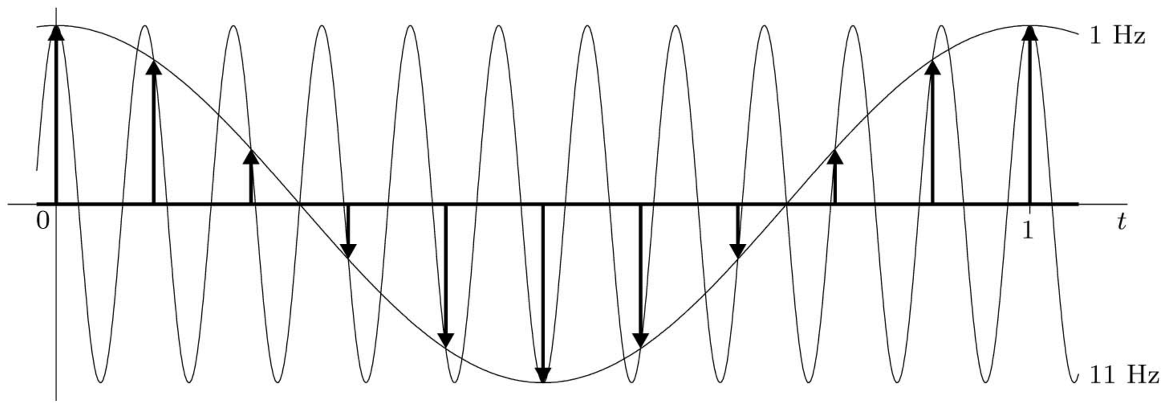
Apparent frequency

$$f_a = |f - m f_s| \qquad f_a < f_s/2$$

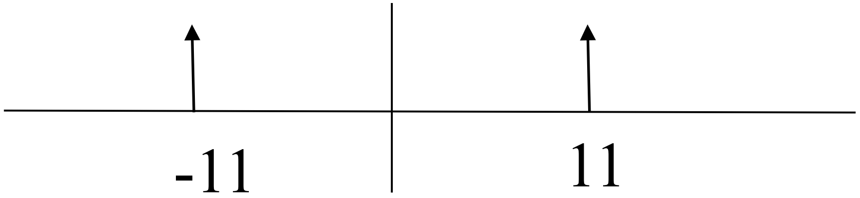
$$f = 3100 \quad f_s = 2000 \quad f_a = ?$$

Aliasing

A sinusoid with 11 Hz frequency sampled with sampling rate $f_s = 10$ z.

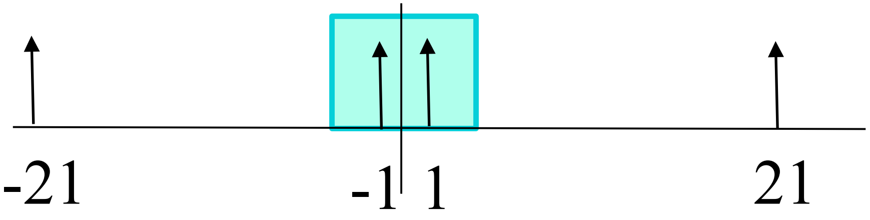


$f_a = |f - mf_s|$ $\quad f_a < f_s/2$



$f = 11 \quad f_s = 10 \quad f_a = ?$

LPF with BW = 5



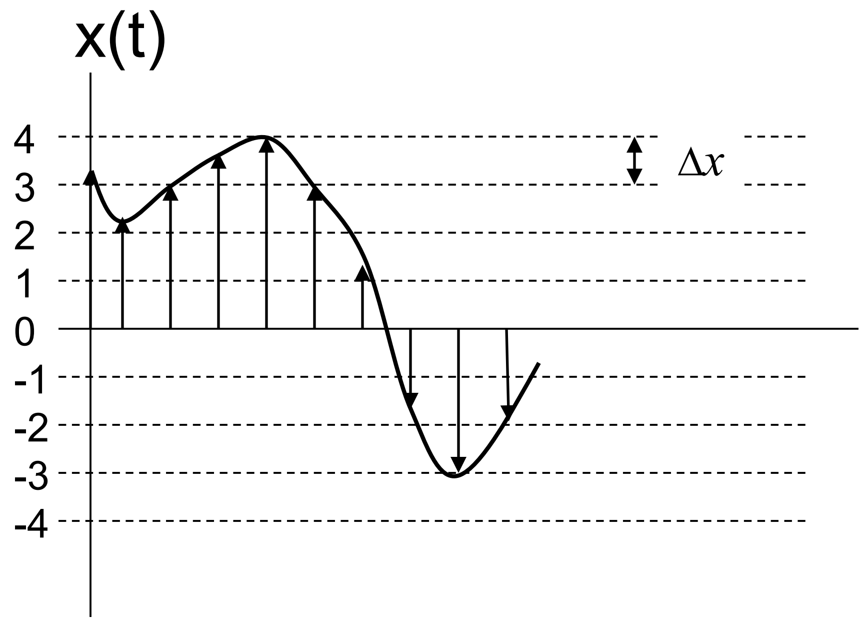
Quantization & Binary Representation

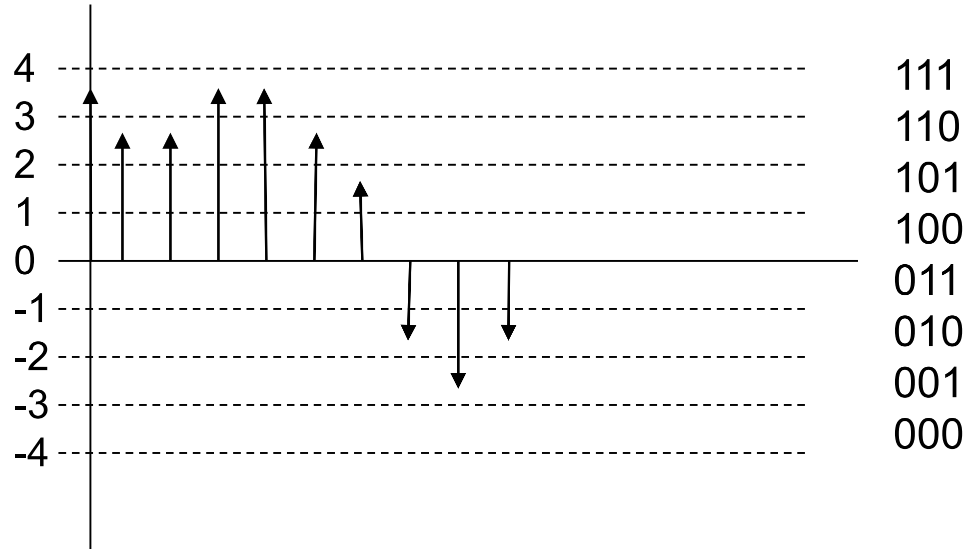
$$L = 2^n$$

L : number of levels
n : Number of bits
Quantization error = $\Delta x/2$

$$\Delta x = \frac{x_{\max} - x_{\min}}{L}$$

$$\Delta x = \frac{2x_{max}}{L}$$





Example

A 5 minutes segment of music sampled at 44000 samples per second. The amplitudes of the samples are quantized to 1024 levels. Determine the size of the segment in bits.

Solution

of bits per sample = ln(1024)   { remember $L=2^n$ }
n = 10 bits per sample
# of bits = 5 * 60 * 44000 * 10 = 13200000 = 13.2 Mbit

Problem 8.3-4

Five telemetry signals, each of bandwidth 1 KHz, are quantized and binary coded, These signals are time-division multiplexed (signal bits interleaved). Choose the number of quantization levels so that the maximum error in the peak signal amplitudes is no greater than 0.2% of the peak signal amplitude. The signal must be sampled at least 20% above the Nyquist rate. Determine the data rate (bits per second) of the multiplexed signal.

Discrete-Time Processing of Continuous-Time Signals

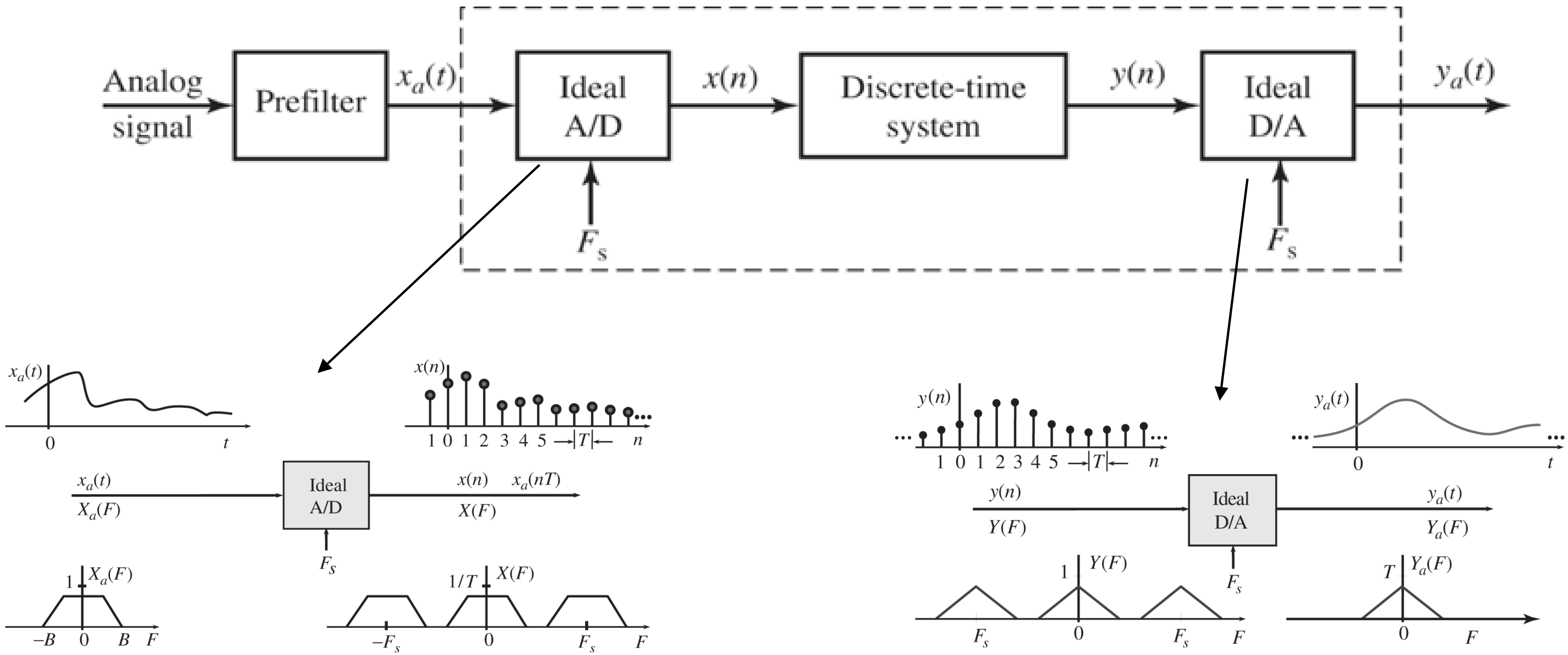


Figure 6.2.2 Characteristics of an ideal A/D converter in the time and frequency domains.

Figure 6.2.3 Characteristics of an ideal D/A converter in the time and frequency domains.

Signals and Systems

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	TQ	T1	T2	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	THW	Attd	Final	Total	Norm	
										25%	15%	15%												15%	10%	20%	80%	100%	
10	5	9.5	9	10	9	8	9	10		24	89	95	20	20	20	19	17	20	20	20	15	18		14	10		76.1	95.1	A
8	8	9.5	9	9	9	7	9	9		24	82	85	16	20	20	17	17	17	15	20	16	18		13	9.8		72.2	90.2	A
9	7	9.5	3	7	10	8	9	10		22	80	90	20	20	20	16	20	20	20	20	15	17		14	9.8		71.6	89.5	A
10	0	10	9	10	10	9	10	10		24	90	81	19	20	20	8	18	17	0	17	18	18		12	9.8		71.2	88.9	A
10	7	9.5	9	10	9	9	5	10		24	76	85	18	20	20	16	15	19	10	19	16	18		13	9.5		70.6	88.2	A/B
8	5	9.5	10	10	5	8	9	8		22	93	87	20	20	20	19	17	18	0	15	9	11		11	9.5		70	87.5	B
7	8	9.5	9	9	9	7	8	9		23	82	87	19	20	20	7	17	17	17	19	16	12		12	9		69.9	87.3	B
10	8	6	4	9	9	5	3	10		20	83	91	17	20	20	17	15	20	20	18	19	18		14	9.8		69.5	86.9	B
6	9	10	9	9	10	7	4	9		22	90	68	18	20	20	17	19	20	10	20	20	18		14	10		69.4	86.8	B
10	6	10	8	10	10	7	4	10		23	79	70	12	20	20	18	20	20	17	19	18	17		14	9.8		69	86.2	B
10	7	8.5	6	9	6	9	5	8		21	76	91	20	20	20	17	18	20	15	19	17	18		14	9		69.2	86.4	B
8	9	9.5	10	8	0	7	4	10		20	83	83	20	20	20	19	20	20	0	17	18	18		13	9.3		67.1	83.9	B
5	8	6	8	8	10	5	5	9		20	85	75	17	20	20	18	18	19	20	19	19	14		14	9.3		66.8	83.5	B
9	7	9.5	7	7	4	8	9	7		21	92	72	17	20	20	14	10	19	0	20	20	18		12	9.5		66.6	83.3	B
8	3	8	7	9	6	8	5	8		19	82	90	6	20	20	7	17	7	15	16	10	15		10	9.3		64.5	80.6	B
8	8	9.5	3	9	10	7	4	9		21	84	52	17	20	20	17	18	19	20	19	18	16		14	9.5		64.2	80.3	B
6	8	0	10	8	10	7	8	10		21	75	89	0	20	20	15	15	7	20	17	0	19		10	7.6		62.7	78.4	B/C
4	7	7	5	4	8	5	4	8		16	67	76	18	20	20	18	18	20	20	19	20	16		14	9.8		61.5	76.8	C
10	7	6	3	6	7	7	3	9		18	78	75	19	20	20	7	15	20	0	20	10	15		11	8.8		60.7	75.8	C
7	5	6.5	3	7	7	5	4	7		16	70	62	19	20	20	20	20	20	20	16	18	15		14	10		59.7	74.6	C
6	8	9.5	3	10	5	6	4	9		19	74	63	18	20	20		12	16	5	20	14	16		11	8.3		58.3	72.8	C
8	7	9	3	9	9	6	4	6		19	74	64	0	20	20	12	12	3	6	10	0	6		6.7	10		56.3	70.4	C
7	5	8.5	0	8	10	8	0	0		14	85	72	18	20	20	0	15	18	17	20	0	0		9.6	8.3		55.9	69.8	C
7	4	8	4	4	9	4	7	7		17	67	52	15	20	20	12	9	13	10	19	14	16		11	9.8		55.4	69.2	C
6	4	8	5	8	6	0	6	9		16	79	52	14	20	20	16	20	19	10	19	18	18		13	6.3		54.8	68.4	C/D
6	8	8	0	8	0	4	4	7		14	60	57	18	20	20	14	20	19	20	18	19	18		14	9		54.6	68.2	C/D
10	4	6	3	4	9	5	3	9		17	60	58	19	20	20	0	10	20	0	20	10	16		10	9.5		53.8	67.3	D
3	4	6	3	5	5	5	3	9		13	70	57	13	20	20	12	17	20	20	19	19	15		13	8		53.5	66.8	D
4	7	8	0	4	3	5	8	6		14	79	57	15	20	20	0	0	19	10	19	12	19		10	7.6		52.2	65.3	D
0	4	9.5	3	0	9	7	0	8		13	68	70	20	20	20	0	10	17	0	10	15	15		9.5	6.1		49	61.3	D
7	7	3	3	0	4	0	4	6		11	64	55	16	20	20	0	0	0	0	19	0	0		5.6	5.6		39.6	49.4	F
3	5	9.5		5	0	0	0	5		8.6	58	60	9	20	20	0	0	0	15	0	0	0		4.8	8		39.1	48.9	F
3	3	0	3	0	5	7	0	0		6.6	52	48	20	20	20	0	16	0	0	16	0	0		6.9	3.2		31.7	39.6	F
0	0	0	0	0	0	0	0	0		0	66	65	0	0	0	0	0	0	0	0	0	0		0	3.9		23.6	29.4	F
0	4	0	0	0	0	0	0	0		1.3	65	0	0	20	20	0	0	0	0	0	0	0		3	1.2		15.3	19.1	F